

Scientific Computing 2026

Week 4: September 21, 23

September 16, 2026

Monday: Nonlinear Equations in 1D

Topics: Intro to Nonlinear Equations
Existence and Uniqueness
Sensitivity and Conditioning
Convergence Rates
Nonlinear Eqs in 1D

Read beforehand: Chapter 5: 5.1–5.4, 5.5.1–5.5.5, 5.5.7

Wednesday: Many-dimensional Nonlinear Equations

Topics: Generalizing to functions $f: \mathbb{R}^n \rightarrow \mathbb{R}^m$.

Read beforehand: Chapter 5.6

Review Questions: 5.1–5.5, 5.10, 5.11

Exercises: 5.1, 5.3, 5.4, 5.10, 5.13